

L1- TripPlannerWithLangChain

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In this tutorial, we are building a simple trip planner using LangChain. The app takes a destination and a few preferences, then outputs a tailored trip plan. We'll use the LangChain Expression Language (LCEL) to chain together components such as prompt templates and LLMs.

```
[1]: # importing the necessary libraries
from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser

from dotenv import load_dotenv
_ = load_dotenv()

[2]: # use GPT-4 to process user input and generate trip plans.
model = ChatOpenAI(model="gpt-4o")

[3]: # use a prompt template to set the context for the LLM to generate personalized
      ↪ trip plans based on user input.

system_template = """
You are a trip planner expert. Help me plan a trip to {destination}.
Consider my preferences for {preferences}.
"""

prompt_template = ChatPromptTemplate.from_messages([
    ('system', system_template),
    ('user', 'What should I do in {destination}?')
])

[4]: # use a simple output parser to extract the LLM's response as a string.
parser = StrOutputParser()

[5]: # now chain the prompt template, model, and output parser to process input and
      ↪ output
trip_planner_chain = prompt_template | model | parser

[6]: # define a function to run a query through the trip planner
def plan_trip(destination, preferences):
```

```
input_data = {"destination": destination, "preferences": preferences}
return trip_planner_chain.invoke(input_data)
```

```
[7]: # helper function to render markdown
from IPython.display import Markdown, display

def render_markdown(md_string):
    display(Markdown(md_string))
```

```
[8]: # run the trip planner with different inputs
result = plan_trip("Paris", "museums, cafes, historical sites")
render_markdown(result)
```

Paris is a city rich in history, art, and culture, so you'll have no shortage of activities to enjoy. Here's a suggested itinerary that focuses on museums, cafes, and historical sites:

0.0.1 Day 1: Explore Historical Sites

- **Morning:**
 - **Eiffel Tower:** Start your trip with a visit to the iconic Eiffel Tower. You can either enjoy the view from the ground or take an elevator ride to the top for a panoramic view of Paris.
 - **Champs-Élysées & Arc de Triomphe:** Stroll down this famous avenue and visit the Arc de Triomphe. Climb to the top for another stunning view of the city.
- **Afternoon:**
 - **Notre-Dame Cathedral:** Visit this Gothic masterpiece. Although it's under restoration, you can still appreciate its grandeur from the outside and explore nearby Île de la Cité.
- **Evening:**
 - **Dinner in the Latin Quarter:** Enjoy dinner in this vibrant area filled with charming cafes and restaurants.

0.0.2 Day 2: Museums and Art

- **Morning:**
 - **Louvre Museum:** Spend your morning at the world-famous Louvre. Don't miss the Mona Lisa and other masterpieces. Consider booking tickets in advance to skip the line.
- **Afternoon:**
 - **Musée d'Orsay:** Head to this museum housed in a former railway station, known for its collection of Impressionist and Post-Impressionist art.
- **Evening:**
 - **Seine River Cruise:** Take a relaxing cruise along the Seine to see Paris's landmarks beautifully illuminated at night.

0.0.3 Day 3: Cafes and Montmartre

- **Morning:**
 - **Breakfast at a Café:** Start your day with a classic Parisian breakfast at a local café. Try Café de Flore or Les Deux Magots for a historic experience.

- **Afternoon:**
 - **Montmartre:** Explore this artistic neighborhood, visit the Basilica of Sacré-Cœur, and wander through Place du Tertre to see artists at work.
- **Evening:**
 - **Dinner in Montmartre:** Enjoy dinner at a local bistro. Consider trying traditional French dishes like coq au vin or duck confit.

0.0.4 Day 4: More Museums and Leisure

- **Morning:**
 - **Musée Rodin:** Discover the works of sculptor Auguste Rodin in this beautiful museum and garden.
- **Afternoon:**
 - **Le Marais:** Explore this historic district known for its narrow medieval streets, trendy boutiques, and cafes. Visit the Picasso Museum if time allows.
- **Evening:**
 - **Dinner at a Rooftop Restaurant:** End your trip with a memorable dinner overlooking the city. Consider Les Ombres or Le Ciel de Paris for stunning views.

0.0.5 Tips:

- **Transport:** Consider purchasing a Paris Visite pass for unlimited travel on public transport.
- **Reservations:** Make reservations for popular attractions and restaurants in advance.
- **Language:** While many people speak English, learning a few basic French phrases can enhance your experience.

Enjoy your trip to Paris!

```
[9]: result = plan_trip("Tokyo", "technology, culture, nightlife")
      render_markdown(result)
```

Tokyo is a vibrant city that offers a diverse range of activities catering to different interests, including technology, culture, and nightlife. Here's a tailored itinerary to help you experience the best of Tokyo:

0.0.6 Day 1: Technology and Innovation

- **Morning:**
 - **Akihabara:** Start your day in the electronics district of Akihabara. Explore multi-story electronics stores like Yodobashi Camera, and immerse yourself in the world of anime and manga with a visit to places like Mandarake.
 - **The National Museum of Emerging Science and Innovation (Miraikan):** Discover the latest in science and technology, including robotics, space exploration, and Earth sciences.
- **Afternoon:**
 - **Odaiba:** Head to this futuristic island with attractions like the teamLab Borderless digital art museum and the life-sized Unicorn Gundam statue at DiverCity Tokyo Plaza.
- **Evening:**

- **Robot Restaurant:** Experience a unique dinner show with robots, lasers, and neon lights. While it’s more about entertainment than cuisine, it’s a must-see for tech enthusiasts.

0.0.7 Day 2: Culture and Tradition

- **Morning:**
 - **Asakusa and Senso-ji Temple:** Visit Tokyo’s oldest temple, and stroll through Nakamise Street for traditional snacks and souvenirs.
- **Afternoon:**
 - **Ueno Park and Museums:** Explore the Tokyo National Museum for Japanese art and artifacts or the National Museum of Western Art. Don’t miss the serene Ueno Zoo.
- **Evening:**
 - **Traditional Kaiseki Dinner:** Enjoy an exquisite multi-course Japanese meal at a traditional restaurant.

0.0.8 Day 3: Nightlife and Entertainment

- **Morning:**
 - **Shibuya and Harajuku:** Start with a visit to the iconic Shibuya Crossing. Head to Harajuku for eclectic fashion, quirky shops, and the Meiji Shrine.
- **Afternoon:**
 - **Shinjuku Gyoen National Garden:** Relax in one of Tokyo’s most beautiful gardens, a perfect contrast to the bustling city.
- **Evening:**
 - **Golden Gai:** Discover this area in Shinjuku known for its narrow alleys filled with tiny, themed bars. Each bar seats only a handful of people, providing a cozy and unique experience.
 - **Roppongi:** For more upscale nightlife, visit Roppongi. It offers a variety of bars, clubs, and international dining options. You might also enjoy the views from Roppongi Hills or Tokyo City View.

0.0.9 Additional Tips:

- **Transportation:** Get a Suica or Pasma card for easy travel on Tokyo’s extensive public transport system.
- **Seasonal Recommendations:** If visiting in spring, enjoy cherry blossoms; in summer, attend a fireworks festival; in autumn, see the vibrant foliage; in winter, enjoy holiday illuminations.
- **Language:** While many Japanese people speak some English, learning a few basic phrases can enhance your experience.

This itinerary should give you a comprehensive taste of Tokyo’s technological wonders, rich cultural heritage, and vibrant nightlife. Enjoy your trip!

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